



Chatbots that win hearts and wallets: Maximizing customer satisfaction and repeat purchases

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ABSTRACT

This study examines how customer satisfaction and their repurchase intentions are shaped by chatbot interactions in online shopping experience. Three human-like chatbot attributes, i.e., responsiveness, anthropomorphism, and social presence, are examined for their influence on customer satisfaction and their repurchase intentions. The moderating influence of e-shopping enjoyment over the relationship between customer satisfaction and their repurchase intentions is assessed. Flow theory is used as an overarching theory to build the research model and explain the results. The data was collected through questionnaire survey from 344 Chinese online shopping participants. We used Structural Equation Modeling (SEM) to analyze the collected data. Findings state that chatbots with responsiveness, anthropomorphism, and social presence are much better equipped to satisfy customer needs. In addition, results show that customer satisfaction affects repurchase intentions. Besides, our findings show that e-shopping enjoyment makes satisfaction even more influential on repurchase intentions. It implies that the affective and attention-demanding aspects of e-commerce play their part in defining a customer's intentions toward repeat purchases. This study will serve as a useful reference for e-commerce platforms that intend to use chatbots to increase customer engagement and loyalty, and marketers who could benefit from their activities.

1. Introduction

The growth of computer-generated conversational shopping assistants, also called chatbots, online customer service representatives, or online chat agents, has been made possible by advancements in artificial intelligence (AI) and information technology (Atienza-Barba et al., 2024; Hui et al., 2024; Méndez-Suárez et al., 2023). These agents are drastically changing the way businesses offer online customer assistance to improve customer-support interactions (Ashfaq et al., 2020). "Software that absorbs natural language as inputs and produces natural language as results, and participates in a chat with the user" is a definition of a virtual conversational assistant (Hui et al., 2024).

Chatbots offer multiple forms of support to online customers across different phases of online shopping (Shahzad et al., 2025). This support usually ranges from pre-transaction to post-transaction. Recent chatbots have capabilities to enhance customers' satisfaction through prompt replies with easy guidance (Ashfaq et al., 2020). Chatbots are also

capable of making conversations with customers to guide them in accomplishing certain tasks, from product inquiry to after-sales queries. For example, chatbots can help customers conducting real-time online purchases, or assist them in searching for diverse products or services on a platform (Lee et al., 2022). Chatbots' support in the decision-making process during the checkout process can help customers make better decisions. In such a way, chatbots can provide personalized recommendations, allowing comparisons between products, and speed up searching for products (Luo et al., 2019; Rodríguez et al., 2024).

Despite having immense potential, integrating chatbots in e-commerce platforms raises some essential questions. Such as, to what extent do the human-like features of a chatbot influence customer interaction and their satisfaction? This study aims to answer this question by examining the collective influence of three chatbot attributes, i.e., responsiveness, anthropomorphism, and social presence on customer satisfaction, which then yields repurchase intentions. Past studies reflect contributions of traditional models like Technology Adoption Model

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(TAM) and Expected Confirmation Theory (ECT) in explaining user perceptions and continuance intention. However, such traditional models often miss affective and immersive dimensions of digital interactions through chatbots. However, in comparison to the traditional model like TAM and ECT, Flow Theory provides an alternative framework for examining psychological absorption and enjoyment, i.e., the dimensions growingly relevant in AI based customer service. As this study is focused on chatbots' humanistic elements, which potentially create an environment that is emotionally and cognitively immersive for online customers, Flow Theory provides a foundation based on which the influence of digital interaction's on customer satisfaction and behavioral intentions can be comprehended.

Past studies have examined chatbot interaction features like responsiveness, anthropomorphism, and social presence etc. (Chen et al., 2021; Klein & Martinez, 2023; Park et al., 2025; Yang et al., 2025). However, all such studies have focused on these chatbot features in isolation, not examining their synergetic effect on customer satisfaction and repurchase intentions. Therefore, the existing literature is only able to provide a fragmented view that is unable to answer how chatbot driven interactions affect customer satisfaction and repurchase intentions, especially in the presence of emotional states such as enjoyment. To address this gap, we have incorporated these elements into a unified framework and have explored their collective influence on customer experience. This integrative and unified approach reflects the novelty of this study. It is unlike previous studies that focused anthropomorphism in isolation, this study examines simultaneous influence of responsiveness, anthropomorphism and social presence on customer satisfaction. Such a synergetic perspective would offer a deeper comprehension of collective influence of multiple human-like features to attain customer satisfaction, a niche that is under explored in the available body of knowledge (Fu et al., 2023; Schanke et al., 2021).

Chatbots are revolutionizing the way business interactions are taking place in today's e-commerce paradigm. This study is inspired by such interactions, and therefore, by keeping e-shopping enjoyment as a boundary condition, it examines the influence of chatbots' responsiveness, anthropomorphism, and social presence on customer satisfaction, which then translates into repurchase intentions. Accordingly, within the Flow Theory framework we examine the aforementioned chatbot attributes to assess if these attributes are capable of producing a deeply engaged and immersive state akin to 'flow' (Csikszentmihalyi, 1975). Past studies have applied Flow Theory to a variety of fields like education, business, and psychology (Fullagar et al., 2013; Lavoie et al., 2022). However, we posit that the Flow Theory can also be relevant for information systems and e-commerce. E-commerce firms can improve their platforms' chatbots with the guidelines drawn from tenets of Flow Theory to provide their customers an immersive e-shopping experience. Based on the core elements of Flow Theory, e-commerce firms can strategize for customer engagement, satisfaction and loyalty enhancement. All this would finally culminate in developing long-term relationships with target customers. The current study supports this archetype by expanding the application of Flow Theory to chatbots' attributes to provide optimal customer experience and satisfaction.

The present study contributes to the body of literature by adding a boundary condition which is e-shopping enjoyment as a moderator of the relationship between customer satisfaction and their repurchase intentions by adding it to the research model. Hence, this study will provide an understanding of the process of chatbot assistance that can result in a realistic and satisfying experience in e-commerce. It could also bring about a change in repurchasing behavior. Therefore, this study's contributions have important practical implications that are not only important but essential in this highly competitive and dynamic e-commerce era. By comprehending the synergetic outcomes of humanistic features like responsiveness, anthropomorphism, and social presence, e-commerce firms will be able to design chatbots that may go beyond the transactional level to more meaningful engagement with customers. Chatbots that are able to create an emotionally touching and

satisfying e-shopping experience can be a critical factor in building customer loyalty and actual repurchase (Holdack et al., 2022).

2. Conceptual background

2.1. Flow theory

Csikszentmihalyi suggested Flow Theory, often known as "optimal experience theory," in the 1970s (Mihaly Csikszentmihalyi, 1975). According to the theory, people experience "flow" when they are involved in an activity that meets their skill level and provides a clear set of goals and fast feedback (Mihaly Csikszentmihalyi, 1975). Flow is described as a heightened state of concentration, focus, enjoyment, and a feeling of control over the activity conducted. The concept states that a balance between the individual's level of skill and the presented challenge of the task can create flow. Individuals may get tired and lose interest if the activity or job is too easy, and the reverse affects them if the task is much beyond their capabilities (Lavoie et al., 2022).

Flow Theory has been widely adopted not only in information systems but also in management, consumer psychology, and business research. Its application has extended into domains such as employee engagement (Fullagar et al., 2013), online marketing environments (Fang et al., 2016), and customer experience design (Holdack et al., 2022). In the current study, Flow Theory serves as a conceptual bridge, linking human-like chatbot features to experiential immersion, satisfaction, and eventual repurchase intentions. Flow has also been linked to several positive effects, including enhanced creativity, productivity, and overall well-being (Biasutti & others, 2011; Ellis, 2018). Although flow theory has garnered a great deal of attention, it presents some complications within the IS context. Bölen et al. (2021) argue that there is debate about how to operationalize flow within IS domains and a lot of ambiguity about such issues as what the main components of the flow experience are, and how different antecedents, dimensions, and consequences of the flow experience relate to one another. It is noteworthy that our paper recognizes these challenges and seeks to add value to the ongoing discourse by adopting a particular conceptualization of flow and applying it to the case of chatbot-customer interactions in e-commerce.

Therefore, the application of Flow Theory in this study further strengthens the novel aspect of this study. By integrating responsiveness, anthropomorphism, and social presence, we create an interaction model that induces a "flow state"—a seamless and engaging customer experience. This holistic perspective is largely missing in prior studies (e.g., Bölen et al., 2021; Fullagar et al., 2013), which have overlooked how the convergence of these features can transform ordinary chatbot interactions into deeply immersive e-commerce experiences (Mihaly Csikszentmihalyi, 1975). Flow theory can be applied for the e-commerce-specific use case of chatbots in particular. It may be interesting to deconstruct the three human factors described above (i.e., responsiveness, anthropomorphism, and social presence) through the lens of Flow Theory.

Flow is also defined by continuity and relevant immediate feedback (Csikszentmihalyi, 1975). Responsiveness can be considered to be a crucial factor in the development of such flow state as it can be the component that helps users to feel a sense of control and to be more immersed in their experience. Furthermore, anthropomorphism can be considered to play a very crucial role in chatbots for creating flow states. By simulating social aspects of human communication, anthropomorphic chatbots can create richer and more enjoyable interactions that can lead to more satisfaction of the necessary conditions for experiencing flow (Lavoie et al., 2022; Schanke et al., 2021). Similarly, social presence can also be seen as a component of flow, as it can be considered to add a sense of relationship to the interaction with a chatbot (Konya-Baumbach et al., 2023). The stronger users feel a social presence from a chatbot, the more immersed they will feel. The relational connection created by social presence in the interaction with

e-commerce chatbots is critical for the experience of satisfaction and flow. Chatbots in e-commerce that are able to simulate a stronger sense of social presence may be able to create a much higher-quality user experience which could lead to increased customer satisfaction and retention.

Thus, Flow Theory provides a valuable framework for understanding how integrating these chatbot features, responsiveness, anthropomorphism, and social presence, can create a deeply immersive and enjoyable customer experience in online shopping environments. By analyzing the role of these elements through the lens of Flow Theory, this study contributes insights into how e-commerce platforms can enhance customer satisfaction and loyalty through chatbot design.

2.2. Chatbot characteristics

Chatbots, the computer programs that are designed to interact with people in a human-like manner using conversation, have been developing rapidly in the past decade and have already begun to change the paradigm of online customer service. The construction of these agents has been steadily improving due to a constant development of AI and Information Technology to create better quality chatbots that can significantly boost a business's customer support interaction (Hui et al., 2024; Vassilakopoulou et al., 2023). In spite of the prevalence of these agents, there has been little research to date on how a combination of more human-like characteristics of a chatbot could possibly predict customer satisfaction and repurchase intentions, which is what the current study aims to investigate and fill the gap. A chatbot is defined as "any software that takes natural language as input and provides natural language as outputs and engages the user in a conversation" (Hui et al., 2024). This simplistic view of these programs does not fully take into account how a combination of more complex and, at first glance, independent design features, such as responsiveness, anthropomorphism, and social presence, has to be harmonized and work in unison to create deeper customer engagement (Ashfaq et al., 2020).

Chatbots help clients at all three stages of online purchase: pre-purchase, purchase, and post-purchase stages. Some of the earliest studies document chatbots' ability to help improve transactional efficiency by simplifying and expediting the product search, comparison, and checkout phases (Lee et al., 2022; Luo et al., 2019). Although these studies are vital for a better understanding of how these agents help improve the purchasing process, they do not account for how more relational elements, such as emotional connection and social presence, may increase satisfaction beyond a functional support (Q. Chen et al., 2023). Moreover, being able to respond immediately is critical but is not nearly enough if this functionality is not combined with a sense of "human-ness" that creates trust and loyalty (Ashfaq et al., 2020).

Incorporating human-like characteristics into chatbots has been argued to be one of the most promising new developments towards increased customer satisfaction with e-commerce support (Fu et al., 2023). However, previous research has yet to look at how the effects of these characteristics may combine to influence customer behavior on the whole. Thus, rather than just looking at the mere presence of these features in a chatbot, the current study models how the synergistic combination of them might influence customer satisfaction and intentions to repurchase from a company. Chatbots, as a relatively new initiative in the field of artificial intelligence techniques and natural language processing methods, alter the form of human-like characters, thus having an impact on the end user's experience in a positive manner (Vassilakopoulou et al., 2023). From this perspective, the present study identified three most critical factors that can affect satisfaction: first, responsiveness, which is referred to as an AI capability of replying to messages in a timely manner (Hui et al., 2024); second, anthropomorphism, the ability of an AI to resemble a human being (Chaturvedi et al., 2023); and lastly, social presence, the ability of an AI to build and maintain social relations with the customers (Baraybar-Fernández et al., 2025; Liu & Yao, 2021).

While numerous studies highlight the importance of individual chatbot characteristics like responsiveness or anthropomorphism in driving customer satisfaction (J.-S. Chen et al., 2021; Fu et al., 2023), the current literature does not sufficiently explore how these elements function together to create a seamless, engaging experience that enhances both satisfaction and repurchase intentions. Several studies state that satisfied customers are more likely to return to the same platform for future purchases, thus increasing customer satisfaction and business success (Chen et al., 2021; Fu et al., 2023). Our study addresses this gap by exploring how the interaction between responsiveness, anthropomorphism, and social presence contributes to a more immersive chatbot interaction, ultimately leading to stronger repurchase intentions. By modeling these relationships, we demonstrate that effective chatbot design goes beyond isolated features and involves creating a synergistic environment where functional and emotional satisfaction converge (Hui et al., 2024; Vassilakopoulou et al., 2023).

3. Hypotheses development

3.1. Responsiveness

Responsiveness of a chatbot is defined as the degree to which the system reacts to users' needs or activities in a prompt, proper and effective manner (Eroglu et al., 2003; Hui et al., 2024). In the context of e-commerce, responsiveness of a chatbot influences the overall user experience and impacts customer satisfaction. The responsiveness of a chatbot is described as its ability to respond to user concerns or interactions in a timely, appropriate, and efficient way (Eroglu et al., 2003; Hui et al., 2024). In the broader context of e-commerce customer interactions, the responsiveness of a chatbot is crucial in shaping the overall user experience and affecting customer satisfaction. Chatbots respond fast, are easy to reach, and can be accessed when required making customers feel welcomed and valuable while also providing them with the delight of interacting with a chatbot (Vassilakopoulou et al., 2023; Xie et al., 2023). In this study, responsiveness also refers to reaction or service efficiency (Chen et al., 2021). Customers view a business to be more technologically advanced and satisfied if its chatbot is efficient and responsive (Cheng et al., 2022). This makes it efficient for the users to interact and is part of Flow Theory's goal of giving users a feeling that they are in control. Additionally, an answering bot that is responsive to customers' needs can address problems much more expeditiously; this makes customers even more satisfied (Ashfaq et al., 2020). There is a complex relationship between responsiveness and satisfaction. Fast responses to problems increase usage levels, expectations about service quality, and trust in the system or business as a whole (Hui et al., 2024). Amid the value-driven e-commerce war, responsiveness is one of the chatbots most important functions in forming good impressions and maintaining long-term connections. Accordingly, this scholarship proposes that the responsiveness of chatbots in the virtual shopping setting will increase the level of satisfaction of the customer with the chatbot, as hypothesized below:

H1: *Responsiveness of the chatbot will increase the level of satisfaction of the customer with the chatbot.*

3.2. Anthropomorphism

Anthropomorphism is described as a technology's proclivity to be interpreted as possessing human traits which include emotions and feelings (Chaturvedi et al., 2023). It refers to how robots may interact with humans in the closest "naturalistic" way it is possible. Conversation is a necessary component of human-machine interaction, which is part of the HCI discipline. To fully benefit from HCI, artificial intelligence-based systems must emulate human interaction processes as well as other aspects of human behavior that have the ability of the human user to social attributions and to elicit associated feelings and behaviors (Schanke et al., 2021; Soomro & Khan, 2025; Zhou et al.,

2023). Anthropomorphic chatbots have a more personalized feel and are therefore more likely to be liked by users (Konya-Baumbach et al., 2023). Treating the machine as more human not only promotes efficient communication methods (Cai et al., 2022; Chaturvedi et al., 2023) but also affords the flow-theory conception of interacting in a relaxed and fun manner that is most acceptable to people (Lavoie et al., 2022). Chatbot anthropomorphism is a key to driving user satisfaction by making the experience of communication more entertaining and compassionate. Only limited scholarships have looked into the impact of anthropomorphism on trust in the setting of systems powered by AI. Waytz et al. (2014), for example, ran an experiment to see if anthropomorphism increased trust in self-driving cars. Fu et al. (2023), investigate the role of anthropomorphism in building trust in chatbots and customer's willingness to use chatbots. According to our study's assumption, anthropomorphic features of chatbots will be more likely to increase the level of satisfaction of the customers with chatbot. As a result, a hypothesis has been put forth:

H2: *The chatbot's anthropomorphism will increase the level of satisfaction of the customer with the chatbot.*

3.3. Social presence

Social presence is connected to the experience of having "personal, sociable, and delicate human contact" through mediated communication (Konya-Baumbach et al., 2023). For instance, it has been observed that incorporating individualized greets into an IT service improves users' perceptions of social presence (Caspi & Blau, 2008). Users' expectations often influence their acceptance of AI-based systems that possess sociable and friendly traits (Liu & Yao, 2021). This can lead to a greater degree of trust among people in this kind of technology (Fu et al., 2023). Interactions that are closer to human-like features will be more satisfying for users. This adds a social dimension that promotes feelings of company and communication, representing the spirit on which Flow Theory rests. The existence of these social elements will raise one's sense of satisfaction (Huang et al., 2017). This kind of socially present chatbot not only offers e-commerce support, it can also improve the quality of interaction and positively affect customer satisfaction. Fu et al. (2023) reported that chatbots stimulate approaches of human warmth, which enhances the trust of the users due to the social presence features of the chatbot. In a similar vein in the context of our study, we claim that human-like chatbots elicit significant sensations of social presence, leading to customer satisfaction. As a result, the following hypothesis is developed:

H3: *The social presence aspect of a chatbot will increase the level of satisfaction of the customer with the chatbot.*

3.4. Customer's satisfaction with chatbot and repurchase intentions

Customer satisfaction is observed as one of the most significant factors of a business's success because it is directly related to a company's competitiveness and profitability (Auh et al., 2011; Cavazotte et al., 2020). Customer satisfaction can lead to a variety of advantages, including improved business standing and reputation, reduced customer turnover, a greater dedication to customer demands during integrated quality planning, reduced marketing costs, transaction costs, and costs associated with good or service failures, and, ultimately, increased employee satisfaction and workforce stability (Bergel et al., 2019; Chen et al., 2023). Given the established importance of customer satisfaction in other fields, in the context of customer satisfaction with chatbot and repurchase intentions is relatively overlooked by past studies. This study contends that customer satisfaction is a key in determining repurchase intention in chatbot's interaction context. When people have a favorable and satisfactory interaction with a chatbot, their entire impression of the business and its services improves (Fu et al., 2023). Satisfied customers have a stronger sense of trust and devotion, making them more inclined to repurchase products or services (F. Li et al., 2021). Friendly

interaction via satisfying chatbot encounters can keep customers interested and work multiple times. As a consequence, user satisfaction with chatbots is a crucial driver of users' repurchases intentions, emphasizing the significance of effortless, efficient, and enjoyable interactions in online settings while emphasizing long-term customer loyalty. Therefore, this study offers the following hypothesis:

H4: *Customer satisfaction with chatbot will be positively related to repurchase intentions in online settings.*

3.5. Moderating role of E-shopping enjoyment

Customers experience many emotions (e.g., anger, enjoyment) when buying online, which influences their recurrent buying habits. Fang et al. (2016) define e-shopping enjoyment as the customer's experience of enjoyment or satisfaction when purchasing online. According to a recent scholarship by Pappas et al. (2016), extremely positive emotions, for instance, pleasure and enjoyment, are essential for reaching a high degree of purchase intention. Enjoyment is one of the most researched pleasant emotions (Chen, 2016; Yang, 2012), and it is thought to play a major role in affecting both engagement duration and repurchase intention in online buying (Tripp et al., 2023). Wankel (1997) defines enjoyment as a good emotion that emerges from satisfying specific intrinsic demands or the cognitive experience of having accomplished something challenging. Flow Theory suggests that when people are involved in an enjoyable activity, they become absorbed in it and thus attains a state of flow and fulfillment. Features of a chatbot assist in creating this fun experience to satisfy the customer, hence leading to high customer satisfaction. This satisfaction then has an impact on the repurchase intentions; hence it acts as an important mediator in the causal flow. As a result, we focused on the moderating function of e-shopping enjoyment in the present investigation.

The moderating effect of the enjoyment of e-shopping is a very important variable in explaining why customer satisfaction with a chatbot leads to repurchase intentions. E-shopping enjoyment functions as a kind of booster, increasing the impact that customer satisfaction makes on generating repurchase intentions (Fang et al., 2016). When users find fulfillment in the shopping experience enabled by a chatbot, the benefit of satisfaction is multiplied (Chen et al., 2023; Kushwaha et al., 2021). It also makes for a better overall user experience and contributes to a more positive reaction toward the chatbot interaction (Schanke et al., 2021). As a result, users tend to turn their satisfaction into stronger repurchase intention. However, twinning e-shopping enjoyment with a moderating influence poignantly points up the significance of making not just satisfactory but also enjoyable relations with chatbots to instill long-standing customer loyalty and repeat business in the world's largest consumer market: e-commerce. A recent study investigated the moderating role of e-shopping enjoyment on the relationship between affective factors and perceived value, which leads to repurchase intentions (Fang et al., 2016). Thus, this study formulates the following hypothesis:

H5: *E-shopping enjoyment will strengthen the positive relationship between customer satisfaction with the chatbot and repurchase intentions.*

3.6. Conceptual model

Fig. 1 shows the research framework of this study.

4. Research methodology

The data involved was gathered from internet customers in China via purposive sampling. The emphasis of this purposive sample was based on users who have formerly interacted with online chatbots or chat assistants for solving their queries. Since the respondents were selected through the non-probability purposive sampling strategy, the sample obtained can be considered as analytically representative but not statistically representative and certain extent of selection and self-selection

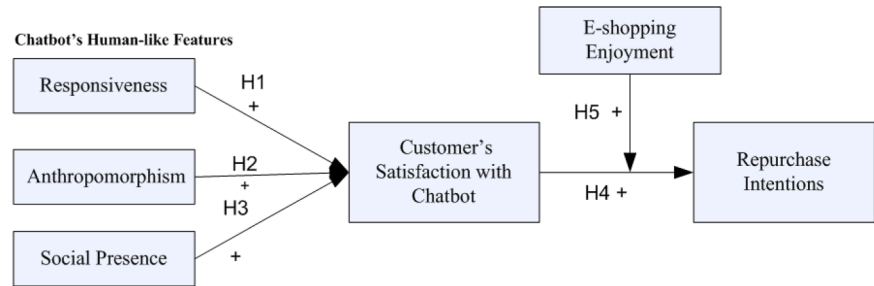


Fig. 1. Research Model.

biasness cannot be eliminated. The Chinese e-commerce business is outstanding, with over 1 billion monthly active users on only three online websites: Taobao, JD.com, and vip.com (Zhu et al., 2023). These and many other e-commerce sites selling to online consumers make use of AI-based mechanisms to service clients. JD.com for example has a built-in AI customer service system that helps customers throughout their purchasing process, including anything from sourcing information and custom-made advice to shipping packages. The Taobao AI endorsement algorithm also stands among the most advanced AI-based e-commerce stores (Zhu et al., 2023). An online survey has been conducted amongst online shoppers, which have previously used AI programs and services when purchasing in China. In order to find potential replies, inclusion criteria have been set. It stipulated that participants were required to be over the age of 18, have an understanding of AI tools offered by e-commerce platforms, and have encountered and used AI tools and services on e-commerce platforms including Taobao and JD suggestion systems, the Alibaba chatbot, Amazon speech recognition solutions, and image search in the previous two months. Screening tests were administered before the main questionnaire based on the aforementioned requirements. Participants who met these requirements were directed to the original survey.

The current research employed a survey administered via the Internet in the Chinese setting, with items written in English being translated into Chinese (Brislin, 1980). Three doctoral students and three professors from universities translated items from English to Chinese autonomously. Experienced linguistic experts then back-translated the questions into English. Disparities in the initial and back-translated versions of the survey were detected and addressed. The final data was gathered via a web-based poll from a select group of online customers utilizing the translated and updated survey. We reduced sample bias by allowing participants to fill out only one response from a single internet protocol (IP) number. We obtained 344 completed surveys for data analysis from 550 online surveys and questionnaires shared on social media, which is deemed a suitable sample size. Our final sample consisted of 344 completed responses, which meets the threshold for structural equation modeling (SEM) analysis. Prior literature suggests that a minimum sample size of 200 is required for models with medium complexity (Kline, 2011), while Hair et al. (2019) recommend 10 respondents per estimated parameter, or at least 300 for robust model estimation. Given the number of constructs and paths in our model, a sample of 344 is considered sufficient and statistically reliable for SEM.

The demographic details of the respondents are listed in Table 1. Among the 344 respondents, the gender distribution was relatively balanced, with 48 % identifying as male and 52 % as female. The majority of participants (28.5 %) were aged between 18 and 24, followed by 27.6 % between 25 and 30, indicating a primarily young adult user base. Most respondents held at least an undergraduate degree (47.7 %), and 36.9 % reported shopping online since last 3–4 years, confirming their relevance for evaluating chatbot-assisted e-commerce interactions.

Table 1
Participant's details.

	Frequency	Percentage
Age in Years		
18–24	98	28.5
25–30	95	27.6
31–35	80	23.3
Above 35	71	20.6
Gender		
Men	165	48.0
Women	179	52.0
Education		
High school diploma	66	19.2
Undergraduate	164	47.7
Masters and above	114	33.1
Online shopper since/for		
Up to 1 Year	35	10.2
1–2 Year	81	23.5
3–4 Year	127	36.9
Above 4 Year	101	29.4

4.1. Measures

The present research used measurements that were already proven to be reliable from previous research. All measures for our constructs were reflective. Reflective measures were used because as the indicators are a result of the presence of a construct and evidence of the consequences of the construct. A five-point Likert scale, ranging from 1-strongly disagree to 5-strongly agree, was used to rate each item as utilized in past studies (Hui & Khan, 2022; Khan et al., 2024; Li et al., 2022). Responsiveness was measured by a 4-item scale adapted from Chen et al. (2021), which has been recently used by Hui et al. (2024). A sample item included was “The AI Chatbot replies quickly”. Anthropomorphism was assessed by a 5-item scale developed by Lu et al. (2019) and recently validated by Fu et al. (2023). One of the sample items was “Chatbots in online shopping environment experience emotions”. Similarly, Fu et al. (2023) used five-item scales to measure the social presence which were adopted from Gefen & Straub (2003), this study utilized the same scale. A sample item was “There is a sense of sociability when communicating with chatbots”. To measure customer satisfaction with chatbot, this study adapted the 4-item scales from Chen et al's (2021) study. A sample item for this factor was “I am satisfied with my overall experience using the chatbot”. E-shopping enjoyment was measured by using a four-item scale adapted from past studies (Chiu et al., 2009; Fang et al., 2016). One of the sample items was “I find visiting this website to be enjoyable”. Finally, to measure the repurchase intentions this study utilized a four-item scale adapted from the existing studies (Fang et al., 2016; Loiacono et al., 2007). A sample item was “If I need to buy a product or service in the future, I would consider buying it from this Website”. Items were slightly modified according to the needs of the current study. Appendix 1 contains all items.

5. Data analysis and results

A range of statistical techniques was employed to investigate the

connections between the aforementioned constructs. Using statistics in SPSS 24.0, the sample's demographic data was compiled. A two-step method for analyzing the proposed model was suggested by Anderson and Gerbing (1988). In line with their methodology, confirmatory factor analysis (CFA) was utilized to ascertain the validity of the measurement items, and as the model was theoretical, structural equation modeling (SEM) was carried out to assess the model fit which has been done recently by past studies (Khan et al., 2026; Li et al., 2026; Li & Khan, 2022; Mehmood et al., 2022). In Amos 24.0, the two stages were completed. The present investigation used CFA rather than exploratory factor analysis (EFA) since all of the latent variables and measurement variables were taken from earlier research that demonstrated their acceptability, validity, and reliability.

5.1. Measurement model

The robust maximum likelihood estimation in CFA was used for evaluating the measurement model. On several goodness-of-fit metrics, the measurement model showed a reasonable fit between the proposed model and the observed data. According to Hair et al.'s (2019) specified benchmark, all goodness-of-fit metrics $\chi^2/df = 1.99$, CFI = 0.95, TLI = 0.94, IFI = 0.95, RMSEA = 0.05, and SRMR = 0.04, fall within the acceptable limits.

Additionally, reliability, discriminant validity, and convergent validity were used to assess the measurement model's suitability. To evaluate convergent validity, we examined the average variance extracted (AVE) for each construct, the significance level for each loading, and the factor loadings for every component in our measuring model. At $p < .01$, every factor loading was statistically significant (See Table 2). In addition, the AVEs significantly fell beyond the suggested threshold of 0.5 (Hair et al., 2019). When combined, these findings gave us sufficient statistical proof that our measuring scales possess convergent validity.

By demonstrating that each construct's square root of AVE was greater than the correlation values among each construct, discriminant validity was determined (See Table 3). All of the construct correlations dropped below the 0.90 criterion, indicating that the constructs are not related to one another. The variance inflation factor (VIF) was calculated to assess multicollinearity. The maximum value of 1.81 reflects the resulting VIF value, and all of them are much below the threshold value of 10. As a result, multicollinearity in our investigation wasn't an issue (Hair et al., 2019). A closer look at reliability was conducted with Cronbach's alpha. Table 2 demonstrates that every construct value in our investigation was higher than the 0.70 acceptability threshold. Harman's single-factor assessment was applied to assess the common method bias (CMB). Six factors with eigenvalues greater than one were found after an exploratory factor analysis was conducted on the items of all latent variables. When combined, these six factors accounted for 75.07 % of the variance in cumulative terms. Less than the permissible 50 % cut-off,

or 37.33 percent, was the amount of variation described by the first factor (Podsakoff et al., 2003). A common latent factor (CLF) was utilized to investigate CMB more profoundly. When this was included in the model, there was no discernible difference in the loadings. These indicated that the study's CMB was not a major problem.

5.2. Structural model analysis and hypothesis testing

The structural model was also compatible with $\chi^2/df = 2.26$, CFI = 0.94, TLI = 0.93, IFI = 0.94, RMSEA = 0.06. After accomplishing the model fit, this study conducted SEM to test the hypotheses. The results of SEM are shown in Fig. 2. Responsiveness (H1: $\beta = 0.24$, $t = 3.80$), anthropomorphism (H2: $\beta = 0.21$, $t = 3.01$), and social presence (H3: $\beta = 0.32$, $t = 4.92$) were significantly and positively related to customer satisfaction with chatbot, stating that H1-H3 all three hypotheses were accepted. Hypothesis 4 states the positive link between customer satisfaction with chatbot and repurchase intentions. This hypothesis was also accepted as this relationship was positive and significant (H4: $\beta = 0.42$, $t = 7.64$).

By calculating and bootstrapping, the standardized path coefficient of the interaction term via SEM, the moderating role of e-shopping enjoyment on the link between customer satisfaction with chatbot and repurchase intentions was tested as proposed in hypothesis 5. The interaction term (e-shopping enjoyment * customer satisfaction with chatbot) on repurchase intentions had a standardized path coefficient of 0.16 ($t = 3.14$). The findings imply that the association between customer satisfaction with chatbots and repurchase intentions is strongly influenced by the enjoyment of e-shopping. The moderating influence of e-shopping enjoyment on the association between customer satisfaction with chatbots and repurchase intentions is seen in Fig. 3. As illustrated in Fig. 3, customer satisfaction with chatbots has a greater impact on repurchase intentions under high e-shopping enjoyment conditions ($\beta = 0.58$, $p < .001$) compared to low e-shopping enjoyment conditions ($\beta = 0.26$, $p < .01$), accepting hypothesis 5 as well.

6. Discussion

Csikszentmihalyi's Flow Theory (1975) has been foundational in explaining immersive experiences; therefore, this study leverages this theory to explore its application in online shopping environments, where customers are drawn into a state of flow when interacting with chatbots. In the context of this study, we used this theory to explain how e-shopping enjoyment serves as a moderator, influencing the relationship between customer satisfaction and repurchase intentions. However, unlike previous applications of Flow Theory, this study integrates multiple human-like chatbot characteristics, exploring their combined effects on customer satisfaction in e-commerce—a gap largely overlooked in the literature. Shopping online puts customers into a “flow” state, and

Table 2
Items loadings, Cronbach's Alpha, Composite reliability (CR), and Average variance extracted (AVE).

Construct	Items	Loadings	Alpha	CR	AVE	Construct	Items	Loadings	Alpha	CR	AVE
Responsiveness (RE)	RE1	0.744	0.888	0.890	0.670	Customer Satisfaction (CS)	CS1	0.831	0.907	0.909	0.714
	RE2	0.842					CS2	0.847			
	RE3	0.802					CS3	0.861			
	RE4	0.880					CS4	0.840			
Anthropomorphism (AN)	AN1	0.813	0.900	0.900	0.645	E-shopping Enjoyment (EE)	EE1	0.802	0.884	0.885	0.659
	AN2	0.770					EE2	0.812			
	AN3	0.914					EE3	0.862			
	AN4	0.782					EE4	0.768			
	AN5	0.725				Repurchase Intentions (RP)	RP1	0.846			
Social Presence (SP)	SP1	0.726	RP2	0.783	0.872		0.874	0.634			
	SP2	0.854	RP3	0.779							
	SP3	0.815	RP4	0.775							
	SP4	0.855									
	SP5	0.847									

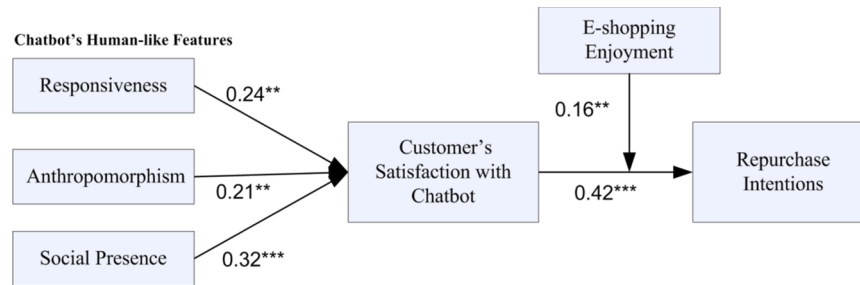
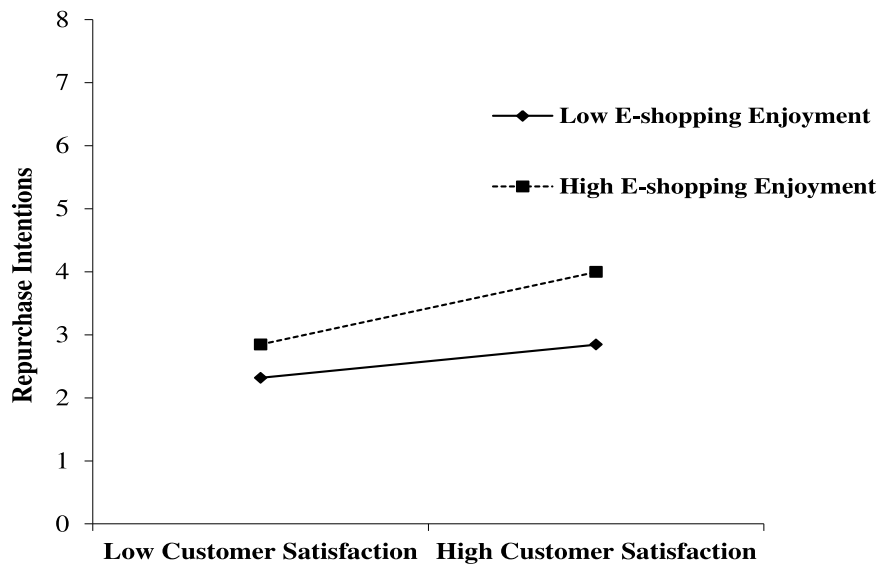
Note. All factor loadings are significant at the $p < .001$ level.

Table 3

Descriptive statistics, square root of AVEs, and correlation matrix.

Constructs	Mean	SD	1	2	3	4	5	6
(1) Responsiveness	3.59	0.82	(0.819)					
(2) Anthropomorphism	3.65	0.81	0.54	(0.803)				
(3) Social Presence	3.60	0.77	0.43	0.59	(0.821)			
(4) Customer Satisfaction	4.01	0.86	0.47	0.51	0.52	(0.845)		
(5) E-shopping Enjoyment	3.68	0.88	0.34	0.26	0.12	0.11	(0.812)	
(6) Repurchase Intentions	3.59	0.90	0.60	0.63	0.49	0.40	0.43	(0.796)

Note: (1) Correlation is significant at the 0.01 level (2) Square roots of AVE for every construct are shown in parentheses.

**Fig. 2.** Path Analysis, Note: *** $p < .01$, ** $p < .01$.**Fig. 3.** Moderating role of e-shopping enjoyment in the relationship between customer satisfaction and repurchase intentions.

they tend to be more open-minded about the effect of satisfaction with chatbots on their repurchase intentions. This is consistent with the concept that being in a state of flow can enhance satisfaction (Mihaly Csikszentmihalyi, 1975), engagement, and involvement to the extent that customers will be more likely to respond positively to good experiences with chatbot interactions.

The study identified three key factors positively related to customer satisfaction with chatbots: responsiveness, anthropomorphism, and social presence. While previous research has studied these factors independently, this study is one of the first to demonstrate their synergistic effects, providing a more integrated understanding of how chatbots can enhance user satisfaction (Fu et al., 2023; Schanke et al., 2021). This integration reveals new insights into how the holistic interaction of these features amplifies customer engagement and satisfaction. What sets this study apart is that it not only reaffirms the importance of these individual characteristics but also shows how they work together to create a seamless, humanized interaction model that enhances the overall

e-commerce experience.

The results demonstrated a positive relationship between customer satisfaction with chatbots and intentions to repurchase. Interestingly, the results showed that e-shopping enjoyment contributes strongly to this relationship; i.e., the more satisfied customers are with chatbots, the stronger their likelihood of repurchasing increases in the presence of e-shopping enjoyment. This insight expands on prior work by revealing that e-shopping enjoyment is not merely a peripheral factor but a critical moderator that significantly enhances the effect of chatbot satisfaction on consumer behavior, especially in the context of repurchase intentions (Holdack et al., 2022; Vassilakopoulou et al., 2023). From this understanding, these findings may be compared with past research. Many studies of e-commerce have always stressed usability, informativeness, cuteness, and hedonic motivation as conditions for satisfying customers (Holdack et al., 2022; Hui et al., 2024; Zhu et al., 2023). However, this study goes beyond these traditional dimensions by highlighting the importance of emotional engagement through chatbot interactions and

demonstrating how the fusion of human-like traits can deepen this engagement, thereby opening new avenues for research on emotional AI in e-commerce.

Additionally, this research contributes to a more effective approach, less studied than the focus on functionalities such as usability, thus bringing a new perspective to the literature. The choice to include anthropomorphism in this study is particularly timely, as previous research had already highlighted how adding human elements in these technological interfaces and encounters can improve the user experience (Hui et al., 2024; Schanke et al., 2021). However, this study builds on these findings by showing that the benefits of anthropomorphism go beyond a direct impact and extend to an enhancement in the effects of the other factors, responsiveness, and social presence. This can be useful for understanding that future research will benefit from further studies to better examine this interplay, for instance in various cultural or technological contexts since this paper's findings have highlighted the contextual nature of customer expectations about human-like experiences (Chaturvedi et al., 2023).

In addition, another surprising and unanticipated discovery disclosed that e-shopping enjoyment had a strong moderating influence on the proposed relationships. While we hypothesized that e-shopping enjoyment would strengthen the association between customer satisfaction and repurchase intentions, the strength of this impact was much higher at the high level of e-shopping enjoyment than anticipated. This result implies that the overall pleasurable experience of the shopping event has an incremental effect on both satisfaction with the chatbots and customers' intention to repurchase. This discovery led to many new and open questions regarding the potential broadened role of emotional states in an e-commerce context. More specifically, it revealed that e-shopping enjoyment has a more central role than previously thought in earlier research (Fang et al., 2016). In addition, it uncovered that the general atmosphere and customers' pleasure experience have a significant influence on the success of implementing such technological advancements as chatbots (Fang et al., 2016).

6.1. Theoretical implications

This study contributes to the literature in several important ways. Theoretically, we provide a more advanced perspective on the role of humanization of artificial intelligence on e-commerce customer satisfaction and repurchase intentions by suggesting the importance of responsiveness, anthropomorphism and social presence. These factors operate in tandem to create a more relatable and human-like interaction with AI systems. In this way, we also expand on current theory regarding TAM by studying the emotive and affective relationship between humans and computers. This investigation is crucial for clarifying and supplementing the existing literature on customer experience and interpersonal interaction in virtual settings, and for redefining and quantifying the patterns of interpersonal interactions in the online world. In this regard, this work highlights the importance of e-shopping enjoyment for decision-making. To date, the relevant research mostly focused on utility (efficiency) and performativity. However, our findings show that the emotional side of customer's experience when shopping online or otherwise in a virtual reality setting – when such experience is 'immersive' and enjoyable – is an important factor that affects satisfaction and customer loyalty. This paper confirms that customer satisfaction and repurchase intention in the e-commerce environment are related not only to product experience but also to the experience of enjoyment and being involved in the shopping process (Fang et al., 2016).

The e-shopping enjoyment as a moderating role is consistent with Flow Theory (Csikszentmihalyi, 1975). That is, our work extends the application of this theory to digital consumer experience. Up until now, this theory was used to explain the level of satisfaction experienced in physical or in activity- or task-oriented contexts. The present study suggests the same is true for virtual environment in which AI agents,

such as chatbots, create such satisfying immersive experiences by facilitating a flow state. A combination of human-like chatbot features and responsiveness to human behavior from chatbots to encourage customers to attain a flow state, and thus increase their level of overall satisfaction in their virtual shopping journey. In this regard, the present study furthers Flow Theory and its application in e-commerce research by suggesting that a flow state is the most critical in e-commerce, as it not only increases satisfaction but also repurchase intentions.

The importance of anthropomorphism in human-computer interaction is also supported by this study. The current research contributes to the anthropomorphism literature by demonstrating that the combination of anthropomorphism, responsiveness, and social presence improve the usability and emotional engagement of this technological system. This study challenges previous user experience research which focuses on the function of a product or service to the exclusion of relational and emotional elements in defining the user experience (Fu et al., 2023). This argument is further reinforced to mean that there is need for change in the design of the man-machine interface in the context of AI so that human-like features can improve user experience and satisfaction with technology (Schanke et al., 2021).

There are also studies that have been published on chatbots or AI's language as the means of communication with customers that we have considered in this current study. The most notable is the study by Jiang et al. (2022), which has placed emphasis on dialogue-based communication through chatbots as the key in the way consumers will perceive their level of satisfaction and engagement in future shopping endeavors. This is aligned with our decision to incorporate the responsiveness of a chatbot, anthropomorphism, and social presence as the drivers of customers' perceived satisfaction. In that light, the So et al. (2023) proposed study on anthropomorphism was concerned with the impact of human likeness on the level of perceived customer satisfaction, but within the limited context of chatbot commerce in the hospitality industry. The conclusion from this study is supported by our results that found anthropomorphic features to improve perceived usability and pleasure (discussed earlier as the dimension of anthropomorphism).

As evident in this study, the results indicated that the emotionality of enjoyment can be explicitly built into the traditional model of customer satisfaction, especially with regard to e-commerce interactions. More specifically, the interplay of emotional and cognitive factors has the implications for a consumer's attitudes and behaviors. In that case, the satisfaction experienced by the consumer does not only come from the product or service being good. It can also come from enjoying an experience that provides a flow to the consumer. This finding, therefore, has contributed to the extension of Flow Theory. It has also called for the revision of the customer satisfaction model to build enjoyment as a source of the repurchase intentions (especially under technology-mediated conditions).

6.2. Practical implications

As can be seen from the above, the results of the study do in fact have practical value, from e-commerce strategy to website design. First, the fact that the chatbot features, i.e., reactivity, anthropomorphism, social presence, influence customer satisfaction in a direct and uncomplicated way has clear and unambiguous guidelines for companies that already use or plan to implement chatbots to assist customers with their requests. Chatbots should be trained to be effective in addressing customer queries but also to provide an experience of humaneness and social presence. This should be the primary goal of organizations as it will greatly increase customer satisfaction. This will also help create a positive image of the brand and could even be a facilitator of customer retention (Chen et al., 2023; Vassilakopoulou et al., 2023).

Second, the fact that customer satisfaction with chatbots has a positive relationship with repurchase intention suggests that the strategic purchase of advanced chatbot technology cannot only be for the convenience of customer service, but for essential methods of attracting

repeat customers. This part of customer contact and retention also becomes more important with increased competition in the online space (Schanke et al., 2021). Third, the buffering role of e-shopping enjoyment in the relationship between chatbot satisfaction and repurchase intention is an important contribution to a well-rounded e-commerce management strategy. This means that a company should attach great importance to providing an enjoyable and interesting e-shopping experience for customers. This can be done by designing a more user-friendly website and providing more personalized recommendations, as well as the use of interactive elements to make the experience “fun”. In this way, organizations supplement the consideration of the end result due to the successful use of chatbots, as it also influences customers to make the decision to re-purchase.

Finally, the unexpectedly strong influence of e-shopping enjoyment invites firms to look beyond chatbot scripts and focus on the wider shopping atmosphere—layout, color, content rhythm, and customer journey coherence. The overall enjoyment customers feel can amplify or dampen the effect of any given technology feature. For decision makers, this means aligning chatbot design with broader experience-management strategies so that technological innovation and emotional engagement reinforce each other across the online platform.

6.3. Limitations and future research

Although this study contributes much to the body of knowledge on the topic, there are also certain limitations to this study that future research can further explore. Relatedly, one such limitation of the study is that the study concentrated on factors like responsiveness, anthropomorphism, and social presence of Chatbots, while not taking account of the other important issues, such as trust, privacy, or security issues, or technology readiness (Cai et al., 2022; Hasan et al., 2021). Additionally, e-shopping enjoyment as a moderating construct, as this study has proposed, may be insufficient to account for the diverse range of motives and behaviors exhibited by different customer segments in e-commerce. Therefore, future research may be conducted to examine such limitations by integrating additional variables that impact customer satisfaction and behavioral intentions with chatbots. For instance, exploring the role of trust, privacy concerns, and perceptions about data security in shaping customers’ attitudes towards chatbots would yield more nuanced insights into the dynamics of human-bot interaction. In this regard, such research could also study how differences in technological readiness and digital literacy levels affect customers’ experiences with chatbots. Such factors might be significant determinants of whether chatbot sessions will be successful for diverse demographic and psychographic segments.

Another possible future research direction is to look at the long-term effects of chatbot conversations on consumer behavior. The present study provides a fair idea of immediate customers’ satisfaction and repurchases intentions. However, knowing the long-term effect that these exchanges have on customer satisfaction or retention can only help to provide a more complete understanding. Moreover, cross-cultural studies of the impact those cultural factors have on customers responses to chatbot traits could reveal interesting findings. How customers perceive and interact with chatbots would be broadly different

across cultures, depending on their peculiar communication styles, technology expectations, and e-commerce habits.

Finally, the study employed a purposive sampling technique (non-probability), targeting participants with prior experience using chatbot-assisted e-commerce platforms. While this approach ensured data relevance and respondent familiarity with the research context, it also may introduce sampling bias and limitations to the generalizability of the findings. The sample may not fully represent the broader population of online shoppers, particularly those with limited or no exposure to chatbots. Future studies could consider using probability-based or stratified random sampling methods to enhance external validity.

6.4. Conclusion

The present study evaluated the impact of chatbot’s human-like qualities (responsiveness, anthropomorphism, and social presence) on customer satisfaction and repurchase intention in e-commerce and confirmed that the interaction of the human-like qualities leads to the increase of customer satisfaction (research question 1: how chatbot’s features can alter consumer’s behavior) and found that e-shopping enjoyment moderated the relationship between customer satisfaction and repurchase intention (research question 2: what is the role of emotion), as confirmed in the literature, had an influence on e-shopping and buying consumers. In this way, the study contributed to the expansion of Flow Theory into virtual AI chatbot-driven environments, as it has shown that the development of immersive, flow-inducing experiences in e-commerce not only heightens customer satisfaction but also has a positive effect on repurchase intentions. This study has also shown that the development of anthropomorphism in interaction will gradually move from interface optimization to the combination of relativity and emotional dimensions, which was another important enlightenment for the future design of AI technology. The study’s results, as a whole, advice that well-known models that are used to account for and measure customer satisfaction should be altered to reflect the integration of cognitive and emotional variables in their influence on consumer behavior.

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CRedit authorship contribution statement

Xiaochao Zhao: Writing – review & editing, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Formal analysis. **Ali Nawaz Khan:** Writing – review & editing, Visualization, Validation, Software, Resources, Methodology, Data curation, Conceptualization. **Khuram Shahzad:** Writing – review & editing, Writing – original draft, Validation, Software, Investigation, Data curation. **Mohsin Ali Soomro:** Writing – review & editing, Writing – original draft, Validation, Resources, Conceptualization.

Appendix 1

Construct	Items
Responsiveness (Chen et al., 2021; Hui et al., 2024)	1 The AI Chatbots replies quickly
	2 Getting in contact with the AI Chatbots is easy
	3 The AI Chatbots is always available when I need it
	4 The AI Chatbots provides credible advice
Anthropomorphism (Lu et al., 2019; Fu et al., 2023)	1 Chatbots in online shopping environment have a mind of their own

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Construct	Items
Social presence (Gefen & Straub, 2003; Fu et al., 2023)	2 Chatbots in online shopping environment have consciousness
	3 Chatbots in online shopping environment have their own free will
	4 Chatbots in online shopping environment experience emotions
	5 Artificially intelligent devices such as robots have intentions
	1 There is a sense of human contact when communicating with Chatbots
Customer satisfaction with Chatbot Chen et al. (2021)	2 There is a sense of personalness when communicating with Chatbots.
	3 There is a sense of sociability when communicating with Chatbots.
	4 There is a sense of human warmth when communicating with Chatbots.
	5 There is a sense of human sensitivity when communicating with Chatbots
	1 I am pleased with using the Chatbot
E-shopping enjoyment (Chiu et al., 2009; Fang et al., 2016)	2 I am satisfied with the pre-purchase experience of using the Chatbot (e.g. product search, quality of information on products or services, product comparison)
	3 I am satisfied with my overall experience using the Chatbot
	4 I would recommend that others use the Chatbot
	1 I find visiting this website to be interesting
	2 I find visiting this website to be entertaining
Repurchase intentions (Fang et al., 2016; Loiacono et al., 2007)	3 I find visiting this website to be enjoyable
	4 I find visiting this website to be pleasant
	1 If I need to buy a product or service in the future, I would consider buying it from this Website
	2 If I need to buy products or service in the future, I would probably revisit this Website
	3 If I need to buy products or service in the future, I would probably try this Website
	4 If I need to buy products or service in the future, I would probably end up making a purchase from this Website

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